PARTIAL FRACTIONS: CASE I AND II

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Name: Solutions

Evaluate the following integrals

1. $\int \frac{x^4}{x-1}\,\mathrm{d}x$

$$
\begin{array}{r}
x^3+x^2+x+1 \\
x-1 \overline{\big) x^4 \qquad\qquad} \\
-x^4+x^3 \\
\hline
x^3 \\
-x^3+x^2 \\
\hline
x^2 \\
-x^2+x \\
\hline
x \\
-x+1 \\
\hline
1
\end{array}
$$

$$\Rightarrow \frac{x^4}{x-1} = x^3+x^2+x+1+\frac{1}{x-1}$$

$$\Rightarrow \int \frac{x^4}{x-1}dx = \int x^3dx + \int x^2dx + \int x\,dx + \int dx + \int \frac{dx}{x-1}$$

$$= \boxed{\frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + \ln|x-1| + C}$$

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2. $\int \frac{5x+1}{(2x+1)(x-1)}\,\mathrm{d}x$

$$\frac{5x+1}{(2x+1)(x-1)} = \frac{A}{2x+1} + \frac{B}{x-1}$$

$$\Rightarrow 5x+1 = A(x-1) + B(2x+1) = (A+2B)x + B-A$$

$$\Rightarrow 5 = A+2B$$

$$1 = B-A \Rightarrow B = 1+A$$

$$\Rightarrow 5 = A+2(1+A) = 3A+2$$

$$\Rightarrow A = 1$$

$$B = 1+1 = 2$$

$$\int \frac{5x+1}{(2x+1)(x-1)}\,dx = \int \frac{1}{2x+1}\,dx + \int \frac{2}{x-1}\,dx$$

$$= \boxed{\frac{1}{2}\ln|2x+1| + 2\ln|x-1| + C}$$

3. $\int \frac{2}{(2x+1)(x+1)}\,\mathrm{d}x$

$$\frac{2}{(2x+1)(x+1)} = \frac{A}{2x+1} + \frac{B}{x+1}$$

$$\Rightarrow 2 = A(x+1) + B(2x+1) = (A+2B)x + (A+B)$$

$$\Rightarrow 0 = A+2B$$
$$2 = A+B$$

$$\Rightarrow 0 = (A+B)+B = 2+B \Rightarrow B = -2$$

$$\Rightarrow 2 = A+(-2) \Rightarrow A = 4.$$

$$\Rightarrow \int \frac{2}{(2x+1)(x+1)}\,dx = \int \frac{4}{2x+1}\,dx + \int \frac{-2}{x+1}\,dx$$

$$= 4\left(\tfrac{1}{2}\ln|2x+1|\right) - 2\ln|x+1| + C$$

$$= \boxed{2\ln|2x+1| - 2\ln|x+1| + C}$$

4. $\int \frac{2x+4}{(x-3)^2}\,\mathrm{d}x$

$$\frac{2x+4}{(x-3)^2} = \frac{A}{x-3} + \frac{B}{(x-3)^2}$$

$$\Rightarrow 2x+4 = A(x-3) + B = Ax + (-3A+B)$$

$$\Rightarrow A = 2$$

$$4 = -3(2) + B = -6 + B \Rightarrow 10 = B$$

$$\Rightarrow \int \frac{2x+4}{(x-3)^2}\,dx = \int \frac{2}{x-3}\,dx + \int \frac{10}{(x-3)^2}\,dx$$

$$= 2\ln|x-3| + 10\left(\frac{-1}{x-3}\right) + C$$

$$= \boxed{2\ln|x-3| - \frac{10}{x-3} + C}$$